Bryan Pham

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. Degree: Mechanical Engineering | GPA: 4.0 / 4.0 — *June. 2026 – June 2028*

Experience

Design & Manufacturing Engineering Intern — June 2026 – September 2026
Mission College | Website — *Santa Clara, CA*

- Designed and optimized the airframe for a high-powered model rocket as a **22-part SolidWorks assembly**, including body tubes, couplers, avionics bay plates, servo mounts, canard shaft bearings, and a fin jig, ensuring structural integrity, steady component implementation, and at least **1.0 cal of stability** in the vehicle
- Conducted **3000+ simulation hours** of computational fluid dynamics (CFD) simulations in SimScale, validating aerodynamic performance, drag, pressure distribution, and stability prior to fabrication
- Engineered an active-control roll system running hardware-in-the-loop at **100 Hz**, streaming **20 Hz IMU** and **10 Hz barometric altitude** over **115200 baud** serial, estimating roll rate with a **Kalman filter** gain-scheduled against fused vertical velocity, and driving two mirrored micro-servo canards clamped to **±7.5°** in flight and **±15°** in firmware
- Prototyped onboard video capture with a Raspberry Pi 5 and Camera Module 3 on a dedicated mount to record canard behavior against CFD-predicted flow, and gated actuation to neutral on any telemetry packet older than **0.5 s**

Co-founder & President of Mission Launch Rocketry — August 2025 – August 2026
Mission College — *Santa Clara, CA*

- Founded and served as President of college rocketry club, overseeing **52 members**, managing budget, coordinating **3 project/promotional events**, and directing design-build-launch projects from concept through flight
- Integrated dual-deployment recovery system in a two-stage high power rocket (drogue and main parachutes) with staged separation sequencing from EasyMini and EasyMega flight computers
- Designed and implemented a minimum diameter airframe in Solidworks for a maximum velocity model rocket, containing an L2 class motor, achieving a simulated **top speed of Mach 2.6**

Projects

Camera Tracking Drone | *Microcomputers, Microcontrollers, Python, Computer Vision* — Code

- Designing toward a 7" autonomous tracking drone that follows only knowingly registered users, starting with a validated Raspberry Pi tracking and CAD design prototype
- Built a real-time **Python** pipeline with CUDA-accelerated **YOLO11n** detection, ByteTrack persistent IDs, and click-to-select targeting emitting a normalized frame-center error signal

RWF DAF Saturation Tank Replacement in Kind | *AutoCAD, Process Mechanical, P&ID, Electrical Design* — Document

- Produced a replacement-in-kind of a dissolved air flotation (DAF) saturation tank at the San José-Santa Clara Regional Wastewater Facility, spanning mechanical, instrumentation, civil, tank, and electrical design
- Drafted **3 of 12** engineering sheets in **AutoCAD: Process Mechanical, P&ID, and Electrical Single-Line** diagrams detailing 6 process streams, 1–8 in. piping, 42 valves, 14 check valves, 4 pumps, and a 48-in.-diameter saturation tank

Aside AI | *1st DeepGram Track, UC Berkeley AI Hackathon | DeepGram, Redis, QNX, Python* — Code

- Built a clip-on camera and mic that narrates the user's surroundings in real time through different AI personalities
- Developed a Raspberry Pi (**QNX 8.0**) capture device, Python coordinator, and React Native app, sending **24 KB** frames every **0.5 s** for **1.8–2.2 s** end-to-end narration

Recco | *Y Combinator AI Growth Hackathon | Computer Vision, Deepgram, Convex* — Code

- Built a camera-first iOS networking assistant that identifies people at events in real time, fusing on-device face tracking with **512-dimensional** InsightFace embeddings and voice control
- Engineered a SwiftUI and AVFoundation pipeline with **Apple Vision face tracking** and an **AR target-lock reticle** that resolves the person nearest screen center on command

Atrium | *1st Overall, Devnovate's Memory Meets Motion Hackathon | TypeScript, Agent Orchestration* — Code

- Built a classroom tool that reads lesson material and student work, groups students by shared misunderstanding, and writes the next lesson around those gaps
- Built a **four-stage** RocketRide pipeline extracting concepts, generating assignment variants, and diagnosing wrong answers, coordinated by **11 schema-validated events** over an **8-node** FalkorDB graph

Technical Skills

Languages : MATLAB, C++, Python, TypeScript, JavaScript, HTML/CSS
CAD: SolidWorks, Onshape, AutoCAD, Fusion360, MecAgent, Zoo.dev
Manufacturing: 3D Printing, Soldering, Microcontrollers, SimScale, Microcomputers, Computer Vision, PSpice, LTSpice, Oscilloscope, Function Generator, Mechanical Systems, Manufacturing Processes, Engineering Drawings, Technical Specifications
Tools & Platforms: Claude Code, Codex, Cursor, Git/Github, Github Actions, Microsoft Office, Redis, Mapbox

Achievements

6x Hackathon Winner: 2nd Overall & 1st Meetstream @ Scalekit x Meetstream, 1st Beginner @ MLH, 2nd Runloop @ Codex